# Lab Exercise 1:

# Introduction to C Programming

Duration: 2 Hours

Topic: Writing, Compiling, Running, and Debugging a Simple C Program

## Learning Outcomes

- Create, write, compile, and run a simple C program.
- Use the printf() function to display text output.
- Identify and correct common syntax errors in C programs.
- Understand the basic structure of a C program.

## Lab Requirements

- Computer with C compiler or IDE (Code::Blocks, Dev-C++, Visual Studio Code with GCC).
- Basic knowledge of file creation and saving in the IDE.

## Part A – Writing, Compiling and Running Your First C Program

1. Task 1: “Hello World” Program

1. Open your C programming IDE.

2. Create a new source file and name it lab1_hello.c.

3. Write the following program:

```c
#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    return 0;
}
```

4. Save, compile, and run your program.

5. Observe the output displayed on the screen.

Expected Output:

Hello, World!

## Part B – Extending the Program

2. Task 2: Adding More Output Statements

Modify your program to display your name, ID, course, and a motivational message using multiple printf() statements.

```
#include <stdio.h>

int main() {
    printf("Hello, World!\n");
    printf("Name   : Ahmad Jamil\n");
    printf("ID     : 123456\n");
    printf("Course : CCC1123 - Programming Concepts in C\n");
    printf("Quote  : Keep learning and never give up!\n");
    return 0;
}
```

## Part C – Identifying Syntax Errors

3. Task 3: Find and Fix the Errors

The following program contains syntax errors. Type it into your IDE, compile it, and correct the errors.

```
#include <stdio.h>

int main()
{
    printf("Welcome to C Programming Lab")
    printf("This is your first program!);
    print("Let's learn together.\n")

    return 0
}
```